

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

CK92425222

I K. Sumanth Kumar Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sumanth

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

- 1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud Computing:-

Cloud computing is the delivery of computing services, storage, databases and analytics over the internet. cloud computing enables business and individuals to use IT resources efficiently without investing in expensive infrastructure.

i) On-Demand Self-Service:

Users can create, modify or remove computing resources whenever required without interacting directly with the cloud provider. This allows organizations to quickly deploy virtual machine, storage whenever needed.

ii) Broad Network Access:

Cloud services are accessible through the internet using standard devices such as desktops, laptops, and tablets. Users can access their applications and data from anywhere with an internet connection.

iii) Resource Pooling:

Cloud providers use virtualization technology to combine computing techniques into shared pool. Multiple customers share the same physical infrastructure while maintaining data security and isolation.

iv) Rapid Elasticity:

cloud resources can automatically scale up or down depending on workload requirement. During high demand additional resources are allocated instantly, and during low demand, unused resources are released.

Conclusion:

cloud computing provides flexibility, scalability, cost savings and global accessibility. These characteristics make it better alternative to traditional on-premise computing for modern organizations.

- 2) Trace the evolution of cloud computing from Hardware evolution to internet software evolution and show how these changes enabled modern cloud platforms
- cloud computing has evolved through several technological developments over many decades.

Initially, organizations used mainframe computers, where multiple users shared a single powerful computer through terminals. Although effective, these systems were expensive and available only to large organizations.

During the 1980s, personal computers (PCs) became popular, allowing individual users to own dedicated

Computer systems. However, organizations had to purchase and maintain separate hardware for every user.

At the same time, the rapid growth of the internet enabled users to access applications and services remotely. High-speed broadband and wireless networks made online computing practical for businesses and individuals.

Software also evolved from desktop applications to web-based applications. Companies began delivering software directly through web browsers using the Software as a Service (SaaS) model.

These developments led to modern cloud platforms that provide infrastructure as a service (IaaS), Platform as a service (PaaS), and Software as a service (SaaS). Cloud providers now offer scalable and cost-effective computing resources through internet.

Conclusion:

The evolution of hardware, virtualization, networking and web technologies laid the foundation for today's cloud computing platforms, enabling organizations to access computing resources anytime and anywhere.

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server Virtualization:

Server virtualization is the process of creating multiple virtual servers on a single physical server using software called a hypervisor. Each virtual machine operates independently with its own operating system and applications.

The major roles of server virtualization include:

- Improving hardware utilization
- Reducing infrastructure and maintenance costs.
- Supporting rapid development of virtual machines.
- Enabling flexible resource allocation.

Virtualization:

Virtualization is an enabling technology that creates virtualized computing resources over the internet. It combines virtualization with automation, networking, security.

⇒ cloud computing as a service model that delivers virtualized computing resources over the internet. It combines virtualization with automation and billing

Systems to provide services on demand.

Virtualization	Cloud computing
i) creates virtual machines	i) Delivers computing services over the internet
ii) Focuses on hardware utilization	ii) focuses on service delivery
iii) can be used without cloud	iii) Uses virtualization as a foundation
iv) managed internally	iv) managed by cloud service providers

Conclusion :

Virtualization enables cloud computing by efficiently managing physical resources, while cloud computing uses virtualization to provide scalable and on-demand IT services.

4) Discuss the role of data centers in delivering cloud services.

A data center is a secure facility containing servers, storage systems, networking equipment, cooling systems, power supplies, and security infrastructure.

The primary role of data center is to host computing resources such as virtual servers, database, storage systems and networking equipment required by cloud customers.

Data centers ensure high availability by using redundant power supplies, backup generators, multiple internet connections, and failover systems. These features keep cloud services available even during hardware failures or power outages.

Data centers also provide security through physical protection such as CCTV surveillance, biometric access control and security personnel. Cybersecurity measures including firewalls, encryption and intrusion detection systems protect customer information from unauthorized access.

Cloud providers use data centers to achieve scalability by adding servers and storage whenever customer demand increases. Virtualization work helps distribute workloads efficiently among available resources.

Conclusion:

Data centers provide secure, reliable, scalable and highly available infrastructure that enables cloud providers to deliver computing services efficiently to users around the world.

5) Differentiate Iaas, Paas, Saas and Xaas with one suitable real-world example for each model.

Cloud computing provides different service models based on the level of responsibility shared between the cloud provider and the customer.

Infrastructure as a service (Iaas):

Iaas provides virtualized computing infrastructure such as virtual machines, storage, networking and operating systems. Customers manage their applications and data, while the provider manages the physical infrastructure. A popular example is Amazon EC2.

Platform as a service (Paas):

Paas provides a complete development platform that allows developers to build, test and deploy applications without managing hardware or operating systems. The cloud provider manages the infrastructure, while developers focus on application development. Google App Engine is a common example.

Software as a service (Saas):

SaaS delivers complete software applications over the internet. Users simply access the software through a web browser without installing or maintaining

it. Google workspace (Gmail, Docs, Drive) is a widely used SaaS application.

Anything as a service (XaaS):

XaaS refers to delivering any IT service through the cloud, including storage, databases, artificial intelligence, security and networking. Dropbox, which provides cloud storage services, is a suitable example of XaaS.

Conclusion:

Each cloud service model offers a different level of management and flexibility. Organizations select the appropriate model based on their business needs, technical expertise and budget.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow